

AMAN DEEP SINGH

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Professional Summary

Backend & Cloud Engineer with 2+ years of experience designing distributed systems, event-driven architectures, and AWS cloud-native infrastructure. Proficient in Django backends, Docker containers, Celery pipelines, and managing client-facing scraping operations to optimize data delivery and incident response.

Certifications

- **AWS Certified Solutions Architect – Associate** (Oct 2025 – Oct 2028) [View Credential](#)
- **AWS Certified CloudOps Engineer – Associate** (May 2026 – May 2029) [View Credential](#)

Experience

SMS DataTech Corporation

July 2023 – Present

Software Engineer (Promoted from Intern)

Tokyo, Japan

- Architected a distributed, LLM-based web scraping platform on AWS ECS/Fargate; engineered dynamic autoscaling policies to process peak loads of 1,000+ concurrent extraction requests without degradation.
- Designed and optimized the Django REST API backend for "TANOMEE" (internal work-sharing and learning suite); reduced database query overhead to lower average API response latency by 45% under peak concurrent loads of 700+ employees. Awarded company-wide MVP (Q4 FY2025).
- Provisioned AWS cloud infrastructure (RDS/Aurora, EC2, S3) and CI/CD pipelines while mentoring junior interns; reduced annual cloud expenditure by executing AWS Savings Plans across 3 projects, saving ~100 USD.
- Created Celery Beat schedulers for periodic data synchronization of the TANOMEE suite with internal tools.
- Designed a self-healing log aggregation and monitoring system with real-time failure alerting during internship; reduced MTTD by 70% and MTTR by 50% for web scraping operations.

Skills

Languages: Python, C/C++, SQL, JavaScript

Frameworks & Tools: Django, Django REST Framework, FastAPI, Scrapy, Celery, Redis, RabbitMQ, Docker

Cloud & DevOps: AWS (EC2, S3, ECS, Fargate, CloudWatch, CloudFormation), Git, GitHub Actions, CI/CD

AI-Driven Development: Codex, Claude Code, Antigravity

Databases: PostgreSQL, MySQL, RDS/Aurora

Concepts: Distributed Systems, Microservices, Event-Driven Architecture, System Design, Scalability, Fault Tolerance, REST APIs

Soft Skills: Leadership, Teamwork, Task Prioritization, Communication, Problem Solving, Mentorship, Adaptability

Projects

Job Hunt AI Pipeline | *Django, MySQL, Celery, Redis, Apify, OpenAI, Discord* | [🔗 Source](#)

- Built an automated, asynchronous pipeline that scrapes job listings, parses them with OpenAI, and ranks them against user profiles.
- Designed a Celery-based workflow with Celery Beat daily triggers, and built a Django REST backend with a dashboard.
- Implemented Redis-based task reconciliation to handle scraping failures and deliver notifications via Discord webhooks.

DistTask | *Python (asyncio), FastAPI, Redis, PostgreSQL, Prometheus, Grafana, Docker* | [🔗 Source](#)

- Developed a lightweight distributed task queue in Python using asyncio and multiprocessing for concurrent task execution.
- Built a worker orchestrator supporting task prefetching, worker heartbeats, exponential backoff retries, and dead-letter queues.
- Tracked queue health, worker statuses, and execution latency using Prometheus metrics and Grafana dashboards.

Education

Indian Institute of Information Technology, Nagpur

2020-2024

Bachelor of Technology in Computer Science and Engineering (CGPA: 8.41/10.0)

Nagpur, India

Achievements & Coding Profiles

- Highest rating of **1455** and rated **Specialist** on [Codeforces](#) (Secured **Global Rank 892** at Round 856 (Div. 2)).
- Solved over **1000 problems** across competitive programming platforms, with active practice on [LeetCode](#).
- Secured State Merit **Rank 10** in the Senior Secondary examination conducted by BSER, Rajasthan in 2019.